

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 6 Lesson 1 Exercise : Introduction to Oracle Application Express

Upload and run a script in APEX

In this exercise you will use the obl Sports file provided for you to create your relational database. This file can be found in Section 6 Lesson 1 (Zip - obl Sports Script) and must first be extracted.

Log on to your APEX environment using your username and password:

1. Click on SQL workshop.
 - a. Click on SQL Scripts.
 - b. Click on upload to browse for and chose the obl Sports.ddl file from your local drive.
 - c. Once you have uploaded the file click on the run button.
 - d. Choose run now
 - e. Click on the view results option and scroll to the bottom of the screen to see if any errors occurred.
 - f. If you had any errors then click on edit script to resolve the errors.
 - g. Rerun the script.

2. Click on SQL workshop.
 - a. Click on SQL commands
 - b. Run the following query to check that the customers table was created properly:

DESCRIBE customers;

- c. Check the description matches your table mapping document.
 - d. Do the same for the other tables that were created.
3. Click on SQL workshop.
 - a. Click on Object Browser to verify the tables through the application instead of through SQL.
 - b. Select tables on the left menu.
 - c. Click on a table to select it.
 - d. Use the tabs along the top to investigate the table information.
4. All changes that you make in APEX are permanent changes so you do not need to save your work.

Task Instructions

Use the completed table mapping document produced in a previous exercise to verify the table creation.

Table Name	Table Short Name			
customers	ctr			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	ctr_number	VARCHAR2	6
uk	*	email	VARCHAR2	50
	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	current_balance	NUMBER	6,2
	o	loyalty_card_number	VARCHAR2	6
fk1	o	tem_id	VARCHAR2	4
fk2	o	sre_id	VARCHAR2	4

Table Name	Table Short Name			
customers_addresses	cas			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	8
	*	address_line_1	VARCHAR2	30
	o	address_line_2	VARCHAR2	30
	*	city	VARCHAR2	15
	*	postal_code	VARCHAR2	7
fk	*	ctr_number	VARCHAR2	6

Table Name	Table Short Name			
teams	tem			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	4
	*	name	VARCHAR2	20
	*	number_of_players	NUMBER	2
	o	discount	NUMBER	2

Table Name	Table Short Name			
sales_representatives	sre			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	4
uk	*	email	VARCHAR2	50
	*	first_name	VARCHAR2	20
	*	last_name	VARCHAR2	30
	*	phone_number	VARCHAR2	11
	*	commission_rate	NUMBER	2
fk	*	supervisor_id	VARCHAR2	4

Table Name	Table Short Name			
sales_representatives_address	sas			
Key Type	Optionality	Column Name	Data Type	Size
Pk, fk	*	id	VARCHAR2	4
	*	address_line_1	VARCHAR2	30
	o	address_line_2	VARCHAR2	30
	*	city	VARCHAR2	15
	*	postal_code	VARCHAR2	7

Table Name	Table Short Name			
orders	odr			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	9
	*	odr_date	DATE	
	*	odr_time	TIMESTAMP	0
	*	number_of_units	NUMBER	2
fk	*	ctr_number	VARCHAR2	6

Table Name	Table Short Name			
ordered_items	oim			
Key Type	Optionality	Column Name	Data Type	Size
	*	quantity_ordered	NUMBER	3
	*	quantity_shipped	NUMBER	3
pk, fk1	*	odr_id	VARCHAR2	9
pk, fk2	*	itm_number	VARCHAR2	10

Table Name	Table Short Name			
items	itm			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	itm_number	VARCHAR2	10
	*	name	VARCHAR2	20
	*	description	VARCHAR2	50
	*	category	VARCHAR2	25
	o	color	VARCHAR2	15
	o	size	CHAR	1
fk	*	ilt_id	VARCHAR2	11

Table Name	Table Short Name			
inventory_list	ilt			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	id	VARCHAR2	11
	*	cost	NUMBER	7,2
	*	units	NUMBER	4

Table Name	Table Short Name			
price_history	phy			
Key Type	Optionality	Column Name	Data Type	Size
pk	*	start_date	DATE	
pk	*	start_time	TIMESTAMP	0
	*	price	NUMBER	7,2
	o	end_date	DATE	
	o	end_time	TIMESTAMP	0
pk, fk1	*	itm_number	VARCHAR2	10